Homework 1: Describing and Comparing Distributions

TEMBA Statistics 2026

Due: See Canvas for your cohort's due date.

Submit a PDF exported from Word. Include every required generated output and your numbered responses, but include only the histogram version you select in Question 9. Round percentages and percentage-point differences to one decimal place.

Each **Generate output** panel asks you to use IntroStatGPT. Use the separate *IntroStatGPT: Homework Workflow* guide for those steps and for exporting your work. Answer each numbered question using the output above it. Some questions also use earlier output, the course notes, or a calculator.

Part 1: Weekly cheese sales and promotional displays

The Kroger cheese data contain 706 weekly observations of packaged, sliced American cheese sales across 11 U.S. cities. The variable `vol` records the number of packages sold, `price` records the retail price per package, and `disp` records whether a promotional display was shown.

Generate output A: Sales histograms

Using the Kroger cheese data:

- Construct histograms of `vol`, one panel for observations with a promotional display and one for observations without one. Put density on the vertical axis, overlay a density curve, and use common horizontal and vertical scales.
- Use **Promotional display** and **No display** as the group labels. Give the plot a descriptive title and clear axis labels.

1. Describe and compare the distributions

Describe each sales-volume distribution. Discuss its location, spread, skewness, modality (number of peaks), and any extreme values. Then describe the two clearest differences between observations with and without a promotional display. Point to evidence in the plot.

2. Interpret the density scale

a. Explain why density rather than count is useful when the two groups have different numbers of observations.
b. Explain what the area in a group's histogram or under its density curve between 3,000 and 5,000 packages represents.

Generate output B: Sales and price summaries

Using the Kroger cheese data:

- Construct one summary table for `vol` with rows for all observations, promotional-display observations, and no-display observations. Give the sample size, missing count, mean, standard deviation, 25th percentile, median, 75th percentile, IQR, minimum, and maximum.
- Construct side-by-side box plots of `vol` for the two display groups on a common axis.
- Construct a second, compact table giving the sample size, mean, median, standard deviation, and IQR of `price` for each display group. Format prices in dollars.
- Use **Promotional display** and **No display** as the group labels. Give every output a descriptive title and clear axis or column labels.

3. Check the sales summary

Use the summary table to confirm that the two display-group counts add to 706 observations. Report the number of missing sales volumes in each group.

4. Compare the centers and spreads

Compare the mean with the median within each display group. Then compare the centers and spreads of the two groups using both the box plots and numerical summaries. If you predicted weekly sales using only display status, which group's sales would be harder to predict? Support your answer with at least one numerical measure of spread.

5. Compare prices

Compare the mean and median price between the two display groups. What does the price comparison add to your interpretation of the sales-volume comparison?

6. Compare the box plots and histograms

Name one feature that is easier to see in the box plots and one that is easier to see in the histograms. What does a point beyond a box-plot whisker mean? Does that alone show that the sales volume is wrong or should be removed? Explain.

7. Apply linear transformations

Use the sales-volume summary table and a calculator.

a. Let $T = V/1000$ express weekly sales in thousands of packages. Convert the median and IQR for each display group to thousands of packages.
b. Let $D = V - \bar{V}$ mean-center sales volume within each display group. State the mean and SD of D for each group.
c. For each transformation, explain what happens to the distribution's shape, center, and spread.

Generate output C: Sales-volume ECDFs

Using the Kroger cheese data:

- Overlay the empirical CDFs of `vol` for the two display groups.
- Format the vertical axis as a cumulative percentage. Use **Promotional display** and **No display** as the group labels, and give the plot clear axis labels.

8. Interpret the ECDF

Use the ECDF to estimate the proportion of each group with weekly sales of at most 5,000 packages. Which group has the larger proportion above 5,000? Explain how you read both answers from the graph.

Generate output D: Histogram fork

Using the visible identifier of your sales histogram:

- Fork the histogram and keep the original.
- Create a second version with narrower bins, using roughly twice as many bins as the original. Leave every other specification unchanged.

9. Compare the bin widths

Identify one feature that changes when the bins become narrower and one overall feature of the distributions that stays about the same. Choose the clearer version for your final PDF and explain your choice.

Part 2: Hybrid work, attrition, and productivity perceptions

Bloom, Han, and Liang conducted a six-month randomized experiment at a Chinese technology company. The 1,612 employees were assigned to a hybrid schedule with two work-from-home days per week or to an in-person control condition.

Employee attrition

The attrition data contain one row per employee. The variable `treat` records the assigned work arrangement. The variable `attrite_perc` equals 100 for an employee who left and 0 for an employee who stayed.

Generate output E: Attrition variable summary

Using the Bloom attrition data:

- Summarize `treat` and `attrite_perc`. Report the software type (such as character or numeric), number of missing values, and distinct observed values for each variable.

10. Classify the attrition variables

State what one row represents. Name the variables for work arrangement and whether the employee stayed or left. Classify `treat` and `attrite_perc` as categorical or quantitative, and state whether each is binary. The software stores `attrite_perc` as a number. Explain why it should still be treated as categorical.

Generate output F: Employee attrition comparison

Using the Bloom attrition data:

- Construct a table by treatment group that gives the group sample size, Stayed and Left counts, and percentages within each group. The variable `attrite_perc` equals 100 for an employee who left and 0 for an employee who stayed.
- Construct a dot plot comparing the two Left proportions on a common percentage axis.
- Use **Control** and **Hybrid WFH** as the treatment labels and **Stayed** and **Left** as the outcome labels.

11. Compare attrition descriptively

Report the Left count and percentage for each group. Explain why percentages are more useful than counts here. Then subtract the Control Left percentage from the Hybrid WFH Left percentage. Report the result in percentage points and interpret it for these 1,612 employees. Use both the table and plot.

Productivity perceptions

The Bloom perceptions data contain employees' responses to a question about how working from home affects productivity. The variable `productivity` keeps the original response labels. Each row gives one employee's response at baseline or endline. We will focus on managers' baseline responses.

> **Generate output G: Perceptions variable summary**
>
> Using the Bloom perceptions data:
>
> - Summarize `wave`, `role`, and `productivity`. Report the software type (such as character or numeric), number of missing values, and number of distinct observed values for each variable.

12. Classify the perceptions variables

State what one row represents. Classify `wave`, `role`, and `productivity` as categorical or quantitative, and state which are binary. For `productivity`, decide whether its categories are nominal or ordinal and explain your answer. Then explain why replacing its labels with the numbers 1 through 9 would not make it a quantitative variable.

> **Generate output H: Managers' perceptions table**
>
> Using the Bloom perceptions data, restricted to managers at baseline:
>
> - Construct a table giving the count and proportion for each original `productivity` response category.
> - Keep the original response labels and place them in their natural order, from the strongest expected productivity loss to the strongest expected productivity gain. Keep that order rather than sorting by frequency.
> - Confirm that the table contains 314 managers.

13. Summarize managers' perceptions

Identify the most common response and report its proportion. Then combine the four "less efficient" responses, the "About the same" response, and the four "more efficient" responses. Report the proportion in each of those three groups.

> **Generate output I: Managers' perceptions plot**
>
> Using the Bloom perceptions data, restricted to managers at baseline:
>
> - Construct a horizontal bar chart or dot plot using proportions.
> - Use the original `productivity` response labels and keep them in their natural order from the strongest expected loss to the strongest expected gain.

14. Interpret the perceptions distribution

Summarize managers' baseline perceptions in two or three sentences. Then explain what the plot would lose if its categories were sorted by frequency.

Optional output J: Compare roles

Using the Bloom perceptions data, restricted to baseline responses from managers and non-managers:

- Produce one plot comparing the two `productivity` distributions on the same proportion scale.
- Use proportions, the original response labels, and the same natural category order for both roles.
- Construct a table giving the negative, about-the-same, and positive proportions for each role.

Optional bonus

Report the negative, about-the-same, and positive proportions for non-managers. Describe the main observed difference between roles.

Export the completed Word document to PDF and submit the PDF through Canvas.